Multilabel-Guided Soft Contrastive Learning for Efficient Earth Observation Pretraining

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***Abstract*— Self-supervised pretraining on large-scale satellite data has raised great interest in building Earth observation (EO) foundation models. However, many important resources beyond pure satellite imagery, such as land-cover-land-use products that provide free global semantic information, as well as vision foundation models that hold strong knowledge of the natural world, are not widely studied. In this work, we show these free additional resources not only help resolve common contrastive learning bottlenecks but also significantly boost the efficiency and effectiveness of EO pretraining. Specifically, we first propose soft contrastive learning (SoftCon) that optimizes cross-scene soft similarity based on land-cover-generated multilabel supervision, naturally solving the issue of multiple positive samples and too strict positive matching in complex scenes. Second, we revisit and explore cross-domain continual pretraining for both multispectral and synthetic aperture radar (SAR) imagery, building efficient EO foundation models from strongest vision models such as DINOv2. Adapting simple weight-initialization and Siamese masking strategies into our SoftCon framework, we demonstrate impressive continual pretraining performance even when the input modalities are not aligned. Without prohibitive training, we produce multispectral and SAR foundation models that achieve significantly better results in 10 out of 11 downstream tasks than most existing SOTA models. For example, our ResNet50/ViT-S achieve 84.8/85.0 linear probing mAP scores on BigEarthNet-10%, which are better than most existing ViT-L models; under the same setting, our ViT-B sets a new record of 86.8 in multispectral, and 82.5 in SAR, the latter even better than many multispectral models. Dataset and models are available at https://github.com/zhu-xlab/softcon.**

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I. Introduction

SELF-SUPERVISED learning has driven wide attention in pretraining Earth observation (EO) foundation models on large-scale satellite data [1], [2], [3]. While more and more efforts are spent on scaling up the data and model size with purely unsupervised pretraining, many other resources such as various land-cover-land-use products tend to be overlooked. For example, ESA WorldCover [4], [5] provides the first global land-cover maps for 2020 and 2021 at 10 m resolution, and Google Dynamic World [6] provides a continuous dataset of near-real-time land-use-land-cover mapping. These dense products are highly correlated with commonly studied medium-resolution satellite imagery and offer free semantic annotations with real-global coverage. Even though they are noisy at pixel-level due to semi-automatic creation process, they can be easily integrated into scene-level annotations with rather good quality. A similar concept has been demonstrated valid in supervised pretraining in GeoKR [7], where land-cover products and geographical location are regarded as geographical knowledge to provide supervision. In this work, we will show the benefits of such auxiliary information in extending the popular contrastive self-supervised learning framework to build EO foundation models.

Contrastive learning with negative sampling such as SimCLR [8] and MoCo [9] has been shown robust and effective in EO pretraining [1], [10]. These methods pull together features of augmented views from the same anchor image (as positive sample) and push away features of other images (as negative samples). The negative samples are usually from a batch or a memory bank, and the model is trained to identify the positive sample from the negative ones. While such instance-discrimination-based methods learn good representations and have been widely used, they inevitably bear the risk of discriminating false negative samples. Specifically, in a large negative pool, there are likely samples that are not the same as the anchor but have very similar semantic information (e.g., belong to the same class). This issue is more significant in EO compared to natural images, as the Earth has a limited surface and the landscapes are usually very redundant.

Several methods have been proposed in computer vision to solve the false negative conflicts, of which one representative is supervised contrastive learning (SupCon) [11]. Leveraging image labels, SupCon defines positive samples as images

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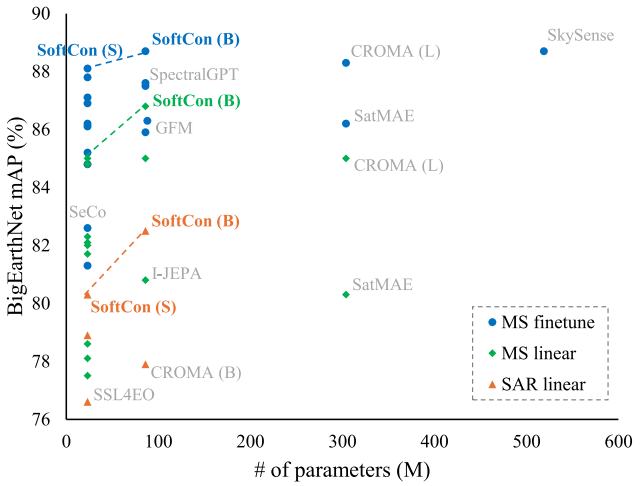


Fig. 1. Visual comparison of transfer learning performances on BigEarthNet-10%. SoftCon (ours) achieves SOTA results with lighter backbones in both linear probing and fine-tuning, as well as in multispectral and SAR. Our best multispectral linear result is comparable to the best models' fine-tuning; our best SAR result outperforms many multispectral models. See Fig. 5 for separated views of each setting.

belonging to the same semantic class. During training, samples from the same class are pulled together in the embedding space, while those from different classes are pushed away. Such a simple multipositive design has been proven beneficial in natural images and, as we mentioned earlier, bears great potential to leverage the free land-use-land-cover annotations in EO. However, a satellite image usually contains more complex semantics than a single class, leading to at least multilabel annotations. Although one can define positive samples as perfect matching of each label component, this would result in two similar images with slightly different label distributions being forced apart (e.g., two neighboring urban scenes one with a small part of river and the other not).

To solve this problem, we extend SupCon to the multilabel scenario by proposing a novel soft contrastive learning method (termed as SoftCon), which takes into account the similarity of complex scenes. Specifically, we calculate cosine similarities of the normalized multilabel one-hot vectors across different scenes. Images with identical labels thus have the highest similarity scores, and images with similar labels have higher scores than images with very different labels. Then, we train the model to directly learn such cross-scene similarities by optimizing a soft contrastive loss on the cosine similarities of corresponding feature projections. In this way, semantically more similar images are pulled closer than semantically more dissimilar images, ending up with a smoothly distributed latent space. To prepare the training data, we match SSL4EO-S12 [10] images with Dynamic World [6] segmentation maps and integrate scene-level multilabel annotations, building a large-scale global multilabel classification dataset.

Meanwhile, another important resource that has huge potential in helping build EO foundation models lies in the general computer vision community: the vision foundation models. These models are exhaustively trained on huge amount of natural images and have already gained strong knowledge of the visual world. Dating back to before the era of foundation models, ImageNet pretrained weights had been widely used and proved beneficial in many supervised EO tasks. Similarly, they can also be used in unsupervised learning, leading to cross-domain continual pretraining. In this regard, recent works such as GFM [12] propose to build EO foundation models by distilling frozen ImageNet models. However, GFM-style training is limited to RGB images, restricting the flexibility to adapt to various EO sensors. Furthermore, the fast advances in computer vision have made available much stronger vision models than ImageNet supervised weights. To bridge this gap, we revisit the natural idea of simple weight initialization, which has been preliminarily explored and verified effective in recent works [13], [14]. For unaligned input modalities, we simply leave the first layer's weights randomly initialized. In addition, to save hardware memory when continually training large Vision Transformers, we adopt Siamese masking inspired by masked autoencoder [15]. Specifically, we randomly mask out a certain percentage of input patches on the trainable branch of the Siamese contrastive learning framework and only send the remaining visible patches to the encoder. We show such simple but flexible continual pretraining strategies, when applied with strong vision foundation models such as DINOv2 [16], exhibit impressive effectiveness and efficiency, even when the imaging sensors are completely different from the source domain (Fig. 1).

Integrating the continual pretraining strategies into the SoftCon framework, we efficiently train CNN and ViT foundation models that reach SOTA performances in 10 out of 11 downstream tasks. For example, our ResNet50/ViT-S (23M parameters, 100 epochs) achieve 84.8/85.0 linear probing mAP scores on BigEarthNet-10%, which are better than most existing ViT-L models (300M parameters, 100–300 epochs) and comparable to CROMA and SkySense ($\geq$600 epochs); under the same setting, our ViT-B (86M parameters) sets a new SOTA of 86.8 in multispectral, and 82.5 in synthetic aperture radar (SAR), the latter even better than most existing multispectral models.

In summary, our contributions are as follows.

1) We explore the benefits of open resources beyond pure satellite imagery for efficient EO pretraining, producing multispectral and SAR foundation models that reach SOTA performances in 10 out of 11 downstream tasks.
2) We propose SoftCon that guides contrastive pretraining with land-cover-generated multilabel supervision.
3) As a side product, we release a global multilabel scene classification dataset by matching noisy Dynamic World land-cover maps with SSL4EO-S12 images and integrating good-quality multilabel annotations.
4) We revisit cross-domain continual pretraining with simple weight initialization from strong vision foundation models and Siamese masking. We verify that even when the input modalities are not aligned, the strong knowledge can still be efficiently transferred to the target EO domain.

II. Related Work

A. EO Foundation Models

Massive research has been conducted on the development of EO foundation models. While there is also a line of supervised

pretraining [17] on large-scale labeled datasets [18], [19], a majority of works tackle the technical adaptation of self-supervised pretraining techniques into EO domain. Early works focus on EO-specific data characteristics for contrastive view generation. For example, Tile2vec [20] proposed to pull together geospatially close tiles while pushing away far tiles. SeCo [21] proposed to use different seasons as augmented views for contrastive learning. CACo [22] proposed to perceive temporal changes with the spatiotemporal structure of remote sensing time series. Further, another group of works explores masked-image-modeling-based pretraining [23]. SatMAE [24] proposed temporal and spectral positional encoding for multispectral imagery and time series. RingMo [25] proposed less aggressive masking. Scale-MAE [26] proposed GSD-based positional encoding and multiscale reconstruction. SpectralGPT [27] proposed 3-D masking to encode and reconstruct spectral data. FG-MAE [28] proposed to reconstruct remote sensing image features such as normalized difference indices. Going beyond a single modality and time stamp, CROMA [29], DeCUR [30], OFA-Net [31], and DOFA [32] investigated multimodal multisensor pretraining, RemoteCLIP [33], SkyScript [34], and BITA [35] explored EO vision language pretraining, RingMo-sense [36], Presto [37], and Prithvi [38] studied EO time series, and SkySense [39] combined both modality and time sequence in a unified architecture, reaching SOTA performance in many downstream tasks. Meanwhile, another line of research targets the curation of EO pretraining datasets, such as SSL4EO-S12 [10] for Sentinel-1 and 2, SSL4EO-L [40] for Landsat series, and SatlasPretrain [41] for medium- and high-resolution satellite and aerial imagery with extensive annotations. While a general trend is to scale the data and model sizes with exhaustive training cost, important existing resources such as open annotations and vision foundation models tend to be overlooked. In this study, we share the conceptual insight with GeoKR [7] to use land-cover-generated multilabel annotations to guide EO pretraining, and with GFM [12] and DOFA to utilize vision models for continual pretraining.

B. Multipositive Contrastive Learning

Contrastive learning beyond simple instance discrimination has been widely explored in computer vision. SupCon [11] proposed to use image labels for positive matching, extending contrastive learning to the fully supervised setting. Almost at the same time, ML-CPC [42] proposed multilabel contrastive predictive coding, identifying multiple positive samples as a multilabel classification problem.[1] Furthermore, WCL [43] combined contrastive instance discrimination with SupCon by predicting pseudo-weak labels. Sel-CL [44] extended SupCon to deal with noisy labels. All of these methods still deal with single-label datasets. To deal with more complex scenes, HiMulCon [45] presented a hierarchical representation learning framework that can leverage all available labels and preserve the hierarchical relationship between classes. MLS [46] proposed to assign multiple binary pseudo-labels for each input image by comparing its embeddings with those in two dictionaries, and training the model with binary cross-entropy loss. However, these two methods are still based on hard multipositive matching, restricting the soft relationship between images with overlapped multilabel distributions. To tackle this issue, and to make the best use of real-world multilabel annotations, we propose SoftCon that allows soft positive matching between different multilabel scenes.

[1]We note that depending on the context, "multilabel" in this article may refer to two things: 1) the contrastive learning objective with multipositive matching and 2) the input images are multilabel annotated.

C. Continual Pretraining

Early works of continual pretraining were mainly developed in natural language processing to improve domain-specific large language models [47], [48]. In vision, existing works like [49] and [50] proposed hierarchical pretraining approaches for task adaptation, while not targeting task-agnostic representations. In remote sensing, CSPT [51] and TOV [52] proposed consecutive pretraining from natural images to remote sensing images, yet they are both limited to retraining on natural images for the source model. SpectralGPT [27] used a similar progressive pretraining pipeline between EO datasets to benefit from their unique advantages, yet it still requires first-stage pretraining from scratch. GFM [12] and DOFA [32] explored continual pretraining for generic EO foundation models by distilling knowledge from frozen vision backbones. However, it heavily relies on ImageNet weights whose prior knowledge is restricted compared to stronger vision foundation models. Meanwhile, GFM-style continual pretraining can only process RGB data, limiting the flexibility to adapt to various EO modalities such as multispectral imagery. To bridge this gap, we revisit and explore the simple cross-domain continual pretraining with weight initialization [13], [14] from strong vision foundation models such as DINOv2 [16]. For non-RGB imagery, we simply leave the input embedding layer randomly initialized. We show such a strategy, though naive, is both flexible and impressively effective in EO pretraining regardless of target sensors.

III. Methodology

A. Building a Large-Scale Multilabel Dataset

We first build a global multilabel land-cover-land-use classification dataset by automatically matching multispectral/SAR imagery from a large-scale satellite dataset with open land-cover products and integrating pixel-level labels to scene-level multilabel annotations.

We choose SSL4EO-S12 [10] as the source of satellite imagery, a multimodal multitemporal dataset specifically designed for large-scale self-supervised learning. The dataset consists of four-seasonal Sentinel-1/2 SAR-optical images from 251 079 non-overlapping locations in the world, covering a wide range of geographical and temporal diversities. For accurate spatial and temporal matching of the scenes, we choose Dynamic World [6] to get the global land-cover maps, which provide continuous land-cover monitoring in nine semantic classes. Both datasets are derived from Sentinel data stored in Google Earth Engine, and thus can be well aligned with the exact metainformation such as the acquisition time.

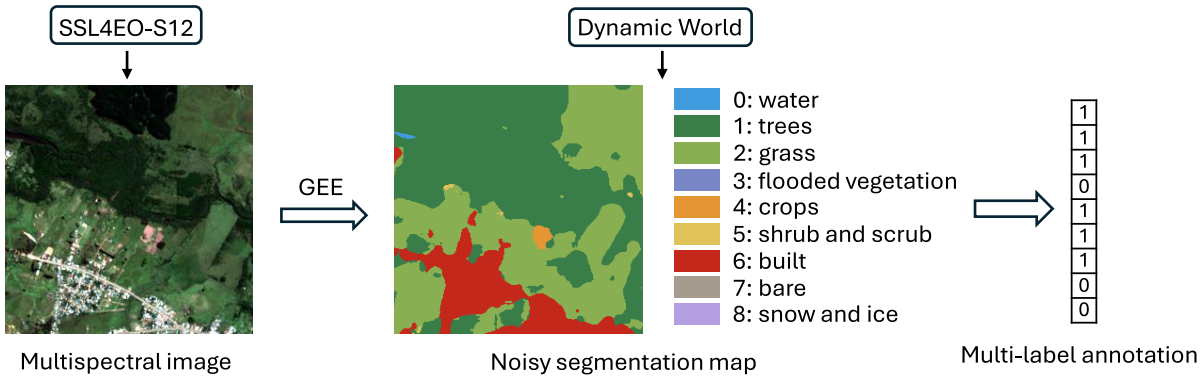


Fig. 2. Workflow of the multilabel dataset curation.

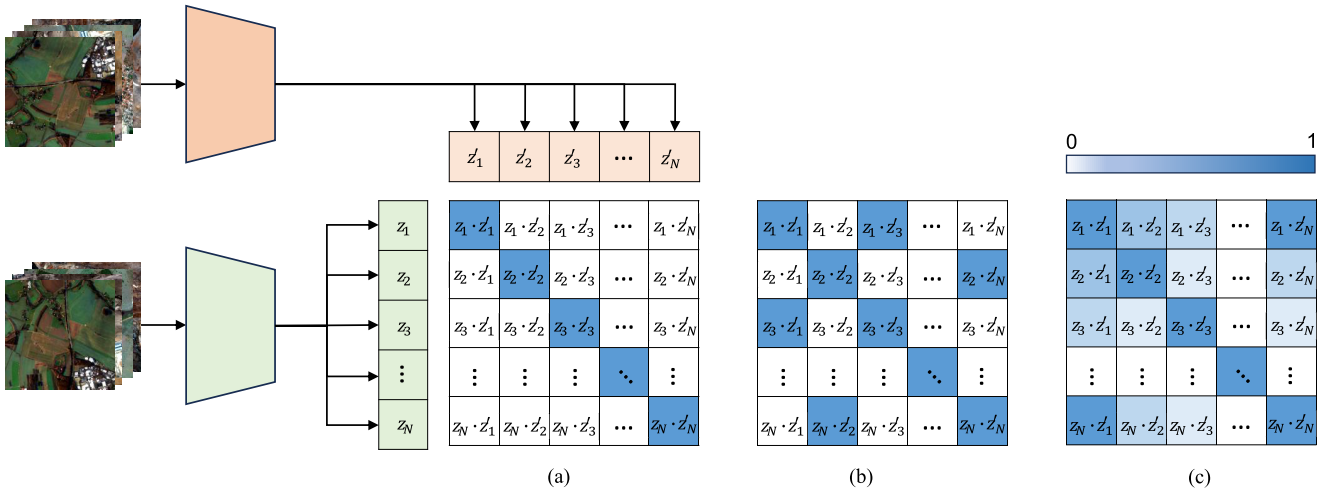


Fig. 3. Different contrastive learning designs. (a) Original contrastive learning performs strict instance discrimination, where each anchor image has only one positive pair. (b) SupCon allows multiple positive responses when images belong to the same class. (c) Our proposed SoftCon can effectively exploit multilabel annotations by assigning soft similarity scores to different pairs. SofCon is a more generic design that covers SupCon: when multilabel degrades to single-label, the soft similarities turn into binary scores, thus becoming multipositive SupCon.

Fig. 2 shows the general workflow of the curation of the multilabel dataset. Based on the geospatial coordinates and acquisition time, we match each SSL4EO-S12 L1C multispectral image with its corresponding Dynamic World land-cover map in Google Earth Engine. Due to the effect of clouds, a few images do not have a corresponding segmentation map. This results in 247 377 locations on which there is a successful match for at least one season. Then, we gather the pixel labels into scene-level multilabels for each image. As can be seen from the example scene in Fig. 2, though the segmentation map is noisy, the scene-level semantic classes are rather accurate.

In total, we get 780 371 annotated multilabel images, each with size 264 × 264 in 10 m resolution. We term this final dataset SSL4EO-S12-ML and will release it for further research. Notably, SSL4EO-S12-ML can also be used as a large-scale multilabel benchmark dataset that covers the whole globe, complementing existing datasets like BigEarthNet [53]. For better reference, we provide a comprehensive dataset sheet and supervised benchmarking results in the Appendix. In addition, we note that SSL4EO-S12-ML labels are derived from Sentinel-2 multispectral images, which are spatially well aligned with the corresponding Sentinel-1 SAR images, but temporally there can be shifts. Therefore, the SAR version is generally less accurate than the multispectral version. Nevertheless, it is enough to guide our pretraining.

B. Multilabel Guided SoftCon

In this section, we introduce SoftCon, which improves upon the original contrastive learning and SupCon [11] to effectively utilize multilabel annotations.

Fig. 3 provides a simplified illustration of the three different contrastive learning designs. Given N raw images, two batches of augmented views are generated and sent through the model to get the corresponding feature vectors $\{z_1, z_2, \ldots, z_N\}$ and $\{z'_1, z'_2, \ldots, z'_N\}$. In the original contrastive learning [Fig. 3(a)], strict instance discrimination is performed such that the cosine similarity of z_i and its augmented view z'_i is maximized, while the cosine similarities of z_i and all other features z'_j $(i \neq j)$ are minimized. This formulates an InfoNCE [54] loss that identifies one positive candidate from all the samples

$$\mathcal{L}^{\text{Contrast}} = -\sum_{i=1}^{N} \log \frac{\exp\left(z_i \cdot z'_i/\tau\right)}{\sum_{j=1}^{N} \exp\left(z_i \cdot z'_j/\tau\right)} \tag{1}$$

where τ is softmax temperature. In practice, contrastive learning typically needs a large number of negative samples. Therefore, it inevitably faces the conflict that multiple images besides the augmented view can belong to the same class as the anchor image while the loss strictly pushes them apart.

SupCon tackles this conflict by introducing image labels to generalize contrastive learning to an arbitrary number of positives. As Fig. 3(b) shows, images belonging to the same class are pulled together, while images belonging to different classes are pushed away. Formally, SupCon accumulates multiple positive pairs while each item has a sole InfoNCE loss

$$\mathcal{L}^{\text{SupCon}} = -\sum_{i=1}^{N} \frac{1}{N_p} \sum_{p \in P(i)} \log \frac{\exp\left(z_i \cdot z'_p/\tau\right)}{\sum_{j=1}^{N} \exp\left(z_i \cdot z'_j/\tau\right)} \tag{2}$$

where $P(i) \equiv \{p \in \{1, 2, .., N\} : y_p = y_i\}$, y is the class label, and N_p is the total number of positive pairs for the anchor feature z_i. SupCon successfully alleviates the single-positive dilemma of the original contrastive learning. However, it only works with single-label imagery and the classes are mutually exclusive.

To effectively exploit multilabel annotations, we propose SoftCon as is shown in Fig. 3(c). Samples with more similar label distributions are pulled closer than samples with more dissimilar labels, resulting in a soft positive matching. Specifically, we calculate the pairwise cosine similarity matrix $Y \in \mathbb{R}^{N \times N}$ of the normalized multihot label vectors $\{y_1, y_2, \ldots, y_N\}$ and $\{y'_1, y'_2, \ldots, y'_N\}$. This is done through the dot product of the label vectors: $\mathbf{y}\mathbf{y}'^{\mathbf{T}}$. Similarly, we get the similarity matrix $X = \mathbf{z}\mathbf{z}'^{\mathbf{T}} \in \mathbb{R}^{N \times N}$ of the feature vectors $\{z_1, z_2, \ldots, z_N\}$ and $\{z'_1, z'_2, \ldots, z'_N\}$. Then, we optimize per-element binary cross-entropy loss

$$\mathcal{L}^{\text{SoftCon}} = -\sum_{i=1}^{N} \sum_{j=1}^{N} \left(Y_{ij} \cdot \log \sigma\left(X_{ij}\right) + \left(1 - Y_{ij}\right) \cdot \log\left(1 - \sigma\left(X_{ij}\right)\right)\right) \tag{3}$$

where Y_{ij} is a soft score between 0 and 1, and $\sigma(\cdot)$ is the sigmoid function.

In practice, we follow MoCo-v2 [55] and MoCo-v3 [56] for the implementation of ResNet and ViT backbones and combine the SoftCon loss with the Contrast loss as

$$\mathcal{L} = \mathcal{L}^{\text{Contrast}} + \lambda \cdot \mathcal{L}^{\text{SoftCon}} \tag{4}$$

where λ is a weighting parameter. This combination is also similarly used by previous works in vision such as [46], [57], [58], and [59]. However, our reason is conceptually different: our nine-class multilabel annotations are on a much coarser level than the real world, thus SoftCon alone may restrict the model's capacity to learn complex landscapes. Also, to prevent potential conflict optimization on the same feature embeddings, we use a separate projector for each objective.

C. Continual Pretraining With Siamese Masking

Finally, we introduce cross-domain continual pretraining into our framework to boost efficient EO pretraining. Instead of sophisticated strategies like GFM [12] that are restricted by RGB input, we revisit the simple weight loading, initializing the backbone with strong vision foundation models. We use DINO [60] weights for ResNet backbones, and DINOv2 [16] weights for ViTs. As for the channel difference between natural RGB images and EO multispectral and SAR imagery, we simply let the input embedding layer remain randomly initialized. Our experiments empirically suggest that though naive, this strategy is both flexible and effective. In addition, to save hardware memory when continually training large vision Transformers, we adopt Siamese masking inspired by iBOT [61], MAE [15], and MSN [62]. Specifically, we randomly mask out a certain percentage of input patches on the trainable branch of the Siamese contrastive learning framework and only send the remaining visible patches to the encoder. As the encoder only needs to process a portion of the full patches, both the memory and the training time can be reduced. Note that we do not conduct any reconstruction like MAE or iBOT, but rather view such masking as additional data augmentation. Different from MSN's non-negative online clustering, we verify that it also works well in contrastive learning with negative sampling: with a reasonable masking ratio of about 20%, not only the efficiency but also the performance can be improved. The full pretraining framework of SoftCon is illustrated in Fig. 4.

IV. Implementation Details

A. Pretraining

We pretrain SoftCon with ResNet [63] and ViT [64] backbones on the integrated multilabel dataset SSL4EO-S12-ML. We normalize the 16-bit multispectral images and the 32-bit SAR images to 8-bit with the mean and standard deviation provided in [10]. If there are multiple seasons for one scene, we randomly choose two for the base encoder and the momentum encoder, respectively. Data augmentations follow [10], including random crop (to the size 224 × 224), color jitter, grayscale, Gaussian blur, and random flip.

We adapt MoCo-v2 [55] for ResNet50 and MoCo-v3 [56] for ViT-small and ViT-base, with two separate projectors to get embeddings for the Contrast and SoftCon loss, respectively. The weighting parameter λ in (2) is 0.1. We set a queue size of 16 384 for MoCo-v2, and a batch size of 1024 for both MoCo-v2 and MoCo-v3. The learning rate is warmed up to $1.5e^{-4}$ for ten epochs followed by cosine decay, and the optimizer is AdamW.

We load ResNet50 weights from DINO[2] [60] and ViT-S/14 and ViT-B/14 weights (without register) from DINOv2[3] [16], and conduct continual pretraining for 100 epochs. For ViTs, we randomly mask out 20% patches and send the remaining ones to the trainable encoder. Training is distributed in two nodes each with four NVIDIA A100 GPUs and takes 7–30 h for different backbones. More details can be seen in the Appendix.

[2]https://github.com/facebookresearch/dino
[3]https://github.com/facebookresearch/dinov2

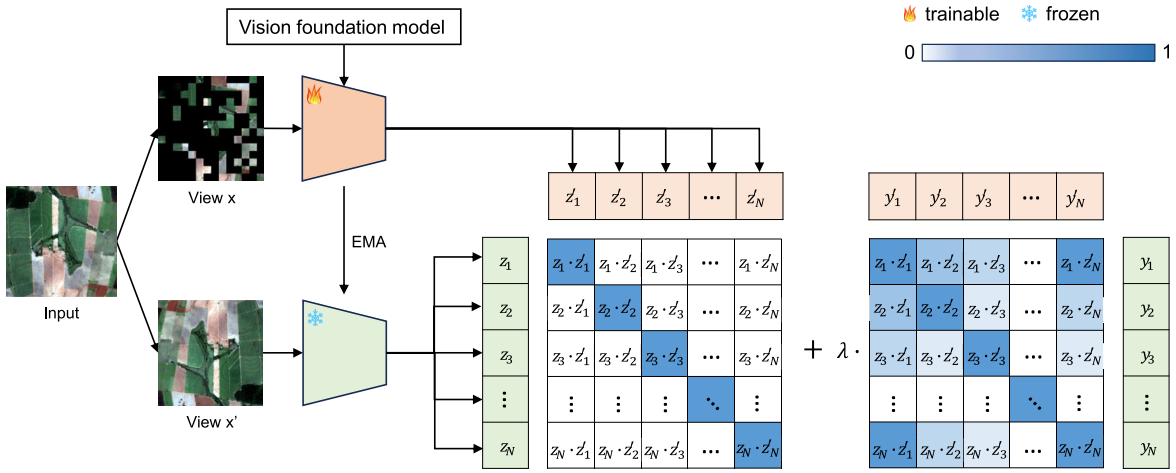


Fig. 4. General framework of SoftCon. Given a batch of input images, two batches of augmented views are parallelly sent through the two branches of a Siamese network. A similarity matrix is calculated based on the resulting two batches of feature vectors. A weighted sum of the contrastive and soft contrastive loss is optimized. For contrastive loss, this matrix should be close to identity; for soft contrastive loss, this matrix should be close to the similarity matrix of the label vectors. We load vision foundation models for both the trainable (base encoder) and the frozen (momentum encoder) branches when initializing the model. During training, the weights of the momentum encoder are updated by exponential moving average (EMA) of the base encoder: $\theta_{\text{momentum}} \leftarrow m \cdot \theta_{\text{momentum}} + (1-m) \cdot \theta_{\text{base}}$, where $m \in [0, 1)$ is a momentum coefficient. Siamese masking is only used with ViT backbones, where random patches of an image are masked out and only the visible patches are sent through the trainable branch.

B. Downstream Tasks

We evaluate the pretrained backbones by linear probing and fine-tuning in 11 downstream tasks, including six land-cover-land-use classification/segmentation datasets: BigEarthNet [53], BigEarthNet-SAR [65], EuroSAT [66], EuroSAT-SAR [28], fMoW-sentinel [24], and DFC2020 [67], one change detection dataset OSCD [68], and four multispectral datasets covering different applications from GEO-Bench [69]: m-so2sat, m-brick-kiln, m-cashew-plantation, and m-SA-crop-type.

1) *BigEarthNet:* A large-scale Sentinel-2 multilabel scene classification dataset covering ten European countries. We use the version of 19 classes and remove bad patches that are fully covered by seasonal snow or clouds. Following previous works [10], [21], [24], [29], [70], we train on 1% or 10% of the training split (31 166 images) and report micro mean average precision (mAP) on the full validation split (103 944 images).
2) *BigEarthNet-SAR:* The paired Sentinel-1 SAR version of the BigEarthNet dataset. We train on 10% of the training split and report mAP scores on the full validation split. The splits are aligned with the above multispectral version.
3) *EuroSAT:* A ten-class scene classification dataset with 27 000 Sentinel-2 images collected from 34 European countries. Following [29], we use 16 200 training images and report overall accuracy on 5400 validation images.
4) *EuroSAT-SAR:* The paired Sentinel-1 SAR version of the EuroSAT dataset.
5) *fMoW-Sentinel:* A large-scale scene classification dataset with 62 classes curated by matching fMoW [71] with Sentinel-2 images. Following [29], we use 10% of the training split (71 287 images) and report top-1 accuracy on the full validation split (84 939 images).
6) *DFC2020:* A ten-class land-cover semantic segmentation dataset that was originally collected for 2020 IEEE data fusion contest. We adjust the official test/validation data with 10 m resolution labels for 5128 training and 986 testing images and report mean intersection over union (mIoU) scores.
7) *OSCD:* A binary change detection dataset consisting of 24 pairs of multispectral images distributed worldwide. We use the official split: 14 training and ten test pairs.
8) *m-so2sat:* A subset of the 17-class So2Sat [72] dataset for local climate zone classification. We use the official split from GEO-Bench with 19992/986/986 train/val/test images and report test top-1 accuracies.
9) *m-brick-kiln:* A subset of the brick-kiln [73] dataset for binary classification. We use the official split from GEO-Bench with 15063/999/999 train/val/test images and report test accuracies.
10) *m-cashew-plantation:* A subset of the cashew-plantation [74] dataset for seven-class semantic segmentation. We use the official split from GEO-Bench with 1350/400/50 train/val/test images and report test mIoU scores.
11) *m-SA-crop-type:* A subset of the SA-crop-type [75] dataset for ten-class semantic segmentation. We use the official split from GEO-Bench with 3000/1000/1000 train/val/test images and report test mIoU scores.

We do a simple grid search for the learning rate for each dataset, using SGD or AdamW optimizer, and train each dataset for 30–100 epochs. We use a common input size 224 × 224 for all datasets. We freeze the encoder and train a linear layer or the decoder for all GEO-Bench datasets. See the Appendix for hyperparameter details.

V. Results

A. Land-Cover Classification

1) BigEarthNet and EuroSAT: We first report SoftCon results on BigEarthNet-10% [53] and EuroSAT [66], which

TABLE I

LINEAR-PROBING/FINE-TUNING RESULTS ON BIGEARTHNET-10% [53] AND EUROSAT [66]. WE REPORT A COMPREHENSIVE COMPARISON WITH SOTA EO FOUNDATION MODELS. MS/SAR/RGB REPRESENT DATA MODALITIES. †: SOFTCON STARTS FROM DINO/DINOV2, WHICH WERE TRAINED ON IMAGENET/LVD-142M. # INDICATES "THE NUMBER OF." *: SKYSENSE EMPLOYS A MIXED ARCHITECTURE (IN TOTAL 2B PARAMETERS) WITH VIT-L AND GEO-CONTEXT ATTENTION TO ENCODE MS IMAGES; 875K STEPS WITH BATCH SIZE 240 ROUGHLY COUNT TO 1000 EPOCHS. LEFT/RIGHT: LINEAR/FINETUNE. BEST SCORES IN **BOLD**

	Pretrain dataset	# Pixels	Epochs	Backbone	# Param.	BE-10%	EuroSAT
Supervised	-	-	-	RN50	23M	83.4	98
MoCo-v2 [10], [55]	SSL4EO-S12 (MS)	70B	100	RN50	23M	82.1/86.2	98.0/99.1
DINO [10], [60]	SSL4EO-S12 (MS)	70B	100	RN50	23M	82.0/87.1	97.2/99.1
SeCo [21]	SeCo (MS)	70B	100	RN50	23M	78.6/82.6	-/93.1
CACo [22]	CACo (MS)	70B	100	RN50	23M	-/81.3	95.9/-
SoftCon (ours)	SSL4EO-S12 (MS)†	70B	100	RN50	23M	**84.8/87.8**	**98.6/99.3**
MoCo-v3 [10], [56]	SSL4EO-S12 (MS)	70B	100	ViT-S	23M	82.3/86.1	97.7/98.6
DINO [10], [60]	SSL4EO-S12 (MS)	70B	100	ViT-S	23M	81.7/86.9	97.7/99.0
MAE [10], [15]	SSL4EO-S12 (MS)	70B	100	ViT-S	23M	77.5/84.8	94.1/98.7
GFM [12]	GeoPile (RGB)	20B	100	Swin-B	88M	-/86.3	-
SpectralGPT [27]	fMoW+BigEarthNet (MS)	12B	300	ViT-B	86M	-/87.5	-/99.2
SatMAE [24]	fMoW (MS)	2.5B	200	ViT-L	304M	80.3/86.2	96.6/99.2
CROMA [29]	SSL4EO-S12 (MS,SAR)	140B	600	ViT-L	304M	85/88.3	**98.0/99.5**
SkySense [39]	SkySense (RGB,MS,SAR)	97T	1000*	ViT-L*	517M*	**-/88.7**	-
SoftCon (ours)	SSL4EO-S12 (MS)†	70B	100	ViT-S	23M	85/88.1	97.1/99.3
SoftCon (ours)	SSL4EO-S12 (MS)†	70B	100	ViT-B	86M	**86.8/88.7**	**98.0/99.5**

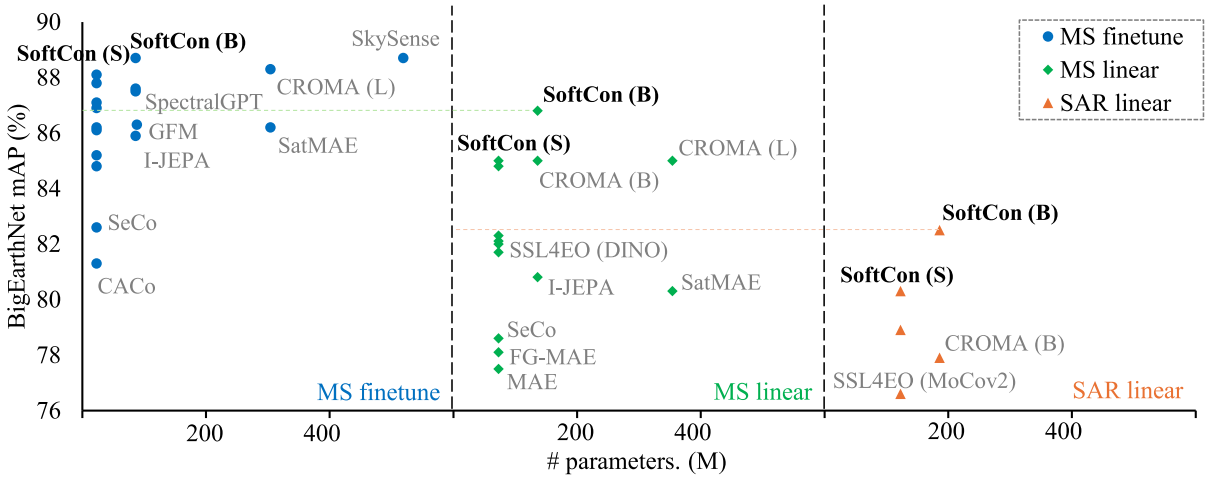


Fig. 5. Detailed comparison of transfer learning performances on BigEarthNet-10%. S/B/L represents ViT-small/base/large. SoftCon (ours) achieves SOTA results with lighter backbones in both linear probing and fine-tuning, as well as in both multispectral and SAR. Our best multispectral linear result is better or comparable to many SOTA models' fine-tuning results; our best SAR result outperforms many multispectral models.

are most commonly evaluated by SOTA EO foundation models. As can be seen from Table I, our models outperform most of the existing works in all scenarios. Specifically on BigEarthNet, our ResNet50 improves over other works with the same backbone by a large margin, with 2.7%/1.6% increase on linear/fine-tuning compared to the current best model, and 6.2%/5.2% increase compared to SeCo [21]. Our linear probing result for the first time outperforms fully supervised training from scratch. Notably, our small ResNet50 also outperforms many existing large ViT models in both linear and fine-tuning, verifying the effectiveness of ConvNet backbones.

For ViT backbones, our ViT-small is already comparable to the best existing models like SpectralGPT [27], CROMA [29], and SkySense [39]. Our ViT-base further pushes forward, achieving a new SOTA of 86.8% mAP on BE-10% with linear probing, and the same performance as SkySense in fine-tuning. More specifically, we achieve the same linear probing performance as CROMA [29] with single-modality pretraining (MS versus MS+SAR), much fewer model parameters (ViT-S versus ViT-L), and much shorter training epochs (100 versus 600). Similarly, we reach the fine-tuning performance of SkySense [39] with the above-mentioned advantages, plus much less pretraining data. For better visual comparison, we provide a scatterplot of various SOTA models' transfer learning performances with respect to model size in Fig. 5.

On EuroSAT [66], the best existing models can achieve a top-1 accuracy close to 99%. This means the dataset is becoming solvable with the fast technological development. As a result, it is less obvious to strictly compare the models' capacities. For example, our ResNet50 is even better than our ViT-base with much fewer parameters. Nevertheless, by comparing both linear and fine-tuning results, our models consistently outperform other works, achieving the same performance as CROMA [29] in both settings.

2) BigEarthNet-SAR and EuroSAT-SAR: Table II reports SoftCon linear probing results in the SAR modality on

TABLE II
LINEAR PROBING RESULTS ON BIGEARTHNET-SAR-10% [65] AND EUROSAT-SAR [28]

	Backbone	BE-SAR-10%	EuroSAT-SAR
MoCo-v2 [10], [55]	RN50	76.6	82.4
SoftCon (ours)	RN50	**78.9**	**87.1**
MAE [10], [15]	ViT-S	69.8	79.3
FG-MAE [28]	ViT-S	71.7	80.7
CROMA [29]	ViT-B	77.9	87.5
SoftCon (ours)	ViT-S	80.3	87.1
SoftCon (ours)	ViT-B	**82.5**	**89.1**

TABLE III
LINEAR PROBING/FINE-TUNING TOP-1 ACCURACY ON FMOW-SENTINEL-10% [24]

	Backbone	Linear	Finetune
DINO [10], [60]	ViT-S/16	32.6	52.8
MAE [10], [15]	ViT-S/16	27.7	51.8
SatMAE [24]	ViT-S/16	35.2	57.2
I-JEPA [76]	ViT-S/16	32.4	53.5
CROMA [29]	ViT-L/8	39.2	59.0
SoftCon (ours)	ViT-S/14	39.9	59.7
SoftCon (ours)	ViT-B/14	**44.0**	**60.6**

BigEarthNet-SAR [65] and EuroSAT-SAR [28] datasets. As we can see, our ResNet50 outperforms MoCo-v2 [55] by 2.3% on BigEarthNet-SAR with 10% labels and by 4.7% on EuroSAT-SAR. Our ViT-S results are much better than MAE [15] and FG-MAE [28] with up to 11.6% improvement. This verifies again the advantage of contrastive learning over masked image model in producing out-of-the-box representations. Our ViT-B sets a new record of 81.4% on BigEarthNet-SAR, 3.5% better than CROMA [29]. *Notably, this score is already higher than many multispectral models* as compared to Table I, achieving a great breakthrough as it has always been very difficult for SAR to beat optical in cloud-free scenes. Our results confirm the great potential of SAR foundation models.

3) fMoW-Sentinel: We present linear probing and fine-tuning results on another more difficult land-cover classification dataset fMoW-sentinel [24] in Table III. As the table shows, our ViT-small already achieves better performance than all existing models. Our ViT-base pushes further, with 4.8% better than the current best model CROMA [29] in linear probing, and 1.6% better in fine-tuning.

B. Land-Cover Segmentation

We report land-cover semantic segmentation results on DFC2020 [67], as shown in Table IV. We fine-tune DeepLabv3+ [77] for ResNet50 and UperNet [78] for ViT backbones. Promisingly, our ResNet50 outperforms MoCo-v2 [55] by 3%, and is also better than existing large ViTs. Our ViT-S outperforms CROMA [29] by 0.9%, which is further improved by our ViT-base with 0.3%. For visual comparison, we plot some example segmentation maps in Fig. 6. As it shows, SoftCon captures more accurate and more fine-grained semantic information compared to SSL4EO-S12 [10].

TABLE IV
FINE-TUNING MIOU RESULTS ON DFC2020 [67]

	Backbone	mIoU
Supervised	RN50	42.9
MoCo-v2 [10], [55]	RN50	47.3
SoftCon (ours)	RN50	**50.3**
MAE [10], [15]	ViT-S	48.0
SatMAE [24]	ViT-L	44.1
CROMA [29]	ViT-L	49.8
SoftCon (ours)	ViT-S	50.7
SoftCon (ours)	ViT-B	**51.0**

TABLE V
RESULTS WITH FROZEN ENCODERS ON OSCD [68]

	Backbone	Precision	Recall	F1
Rand. Init.	RN50	72.3	13.8	23.1
SeCo [21]	RN50	**74.9**	17.5	28.3
SSL4EO [10]	RN50	70.2	23.4	35.1
SoftCon (ours)	RN50	66.6	**29.2**	**40.6**

C. Change Detection

We report binary change detection results on OSCD [68], as shown in Table V. We freeze the backbone and fine-tune a simple U-Net [79] for segmentation. The differences in feature maps between two timestamps are input to the network. SoftCon outperforms SeCo and SSL4EO in both recall and $F1$-score. The low precision score is due to the significant class unbalance, i.e., predicting all pixels as unchanged would result in a good precision score. The combined $F1$-score highlights the superior performance of SoftCon.

D. Other Domain-Specific Applications

Finally, we evaluate SoftCon on four other domain-specific applications with the corresponding Sentinel-2 dataset collections from GEO-Bench [69]. These include two classification datasets m-so2sat and m-brick-kiln, and two segmentation datasets m-cashew-plantation and m-SA-crop-type. We conduct frozen-encoder training following [31], with a linear classifier for classification tasks and UperNet decoder for segmentation tasks. The results are shown in Table VI. We compare SoftCon with CROMA [29] and a recent work OFA-Net [31] and report the official fine-tuning (from timm [80]) results in [69] for reference. The table shows that SoftCon significantly outperforms both CROMA and OFA-Net on three tasks, while only slightly worse than OFA-Net on m-SA-crop-type. Notably, our results with frozen-encoder outperform the official results with full fine-tuning on the two segmentation datasets, and only slightly worse on m-so2sat.

VI. ABLATION AND DISCUSSION

For all ablation studies, we conduct linear probing experiments on BigEarthNet [53].

1) SoftCon Loss: We ablate ResNet50 results on the soft contrastive loss in (4), which are shown in Table VII. In line with our explanation in Section III-B, SoftCon alone is worse than Contrast alone since the ten Dynamic World classes are too coarse-grained to fully represent the real-world semantics. However, combining the two losses provides significant

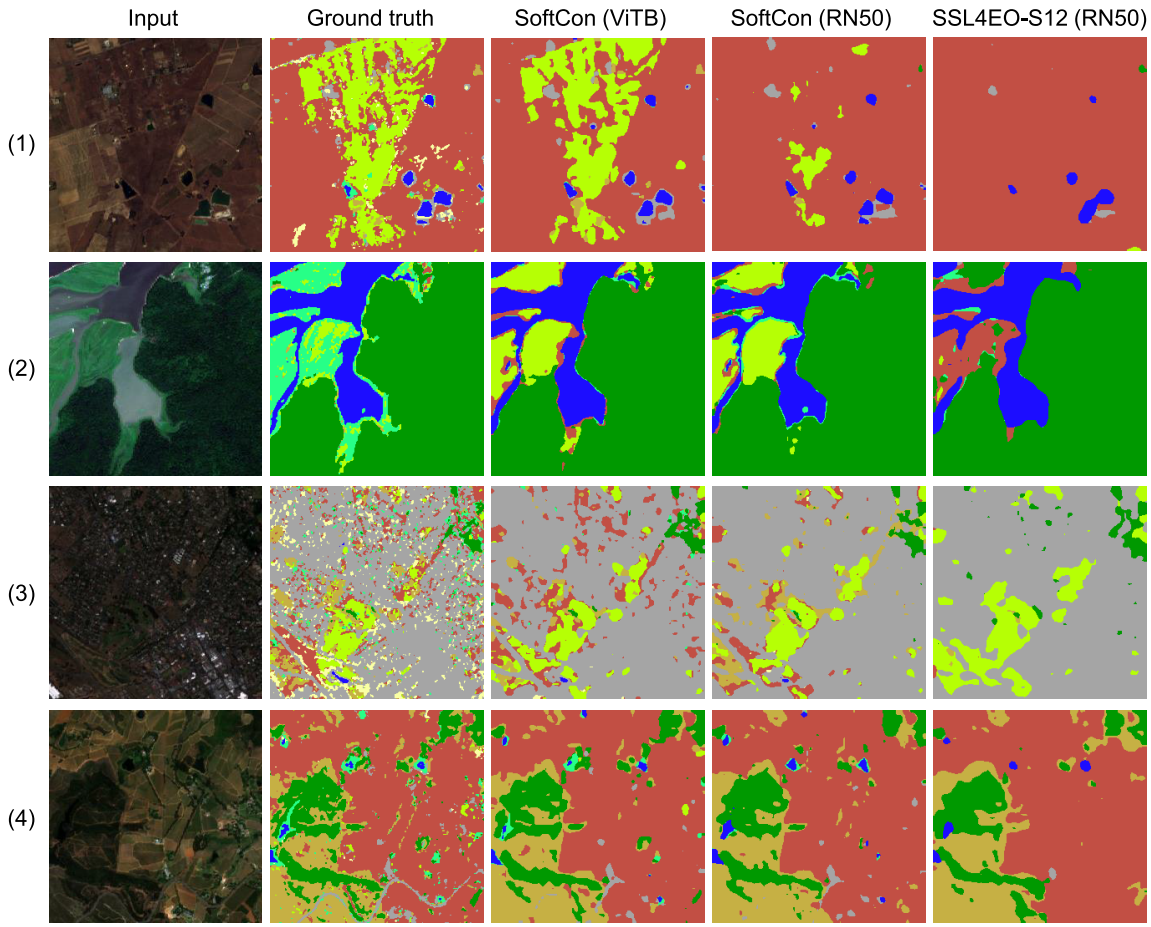


Fig. 6. Example segmentation maps on the DFC2020 [67] dataset.

TABLE VI

TRANSFER RESULTS WITH FROZEN ENCODERS ON FOUR SENTINEL-2 TASKS IN GEO-BENCH [69]. WE REPORT TOP-1 ACCURACY/MIOU FOR CLASSIFICATION/SEGMENTATION, RESPECTIVELY

		m-so2sat	m-brick-kiln	m-cashew-plantation	m-SA-crop-type
GEO-Bench (FT)	RN50	52.8	98.7	44.7	29.9
CROMA [29]	ViT-B	49.2	91.0	-	31.4
OFA-Net [31]	ViT-B	46.0	91.3	37.4	**32.0**
SoftCon (ours)	ViT-S	49.9	92.6	44.4	31.5
SoftCon (ours)	ViT-B	**52.0**	**95.2**	**49.6**	31.5

benefits. When we degrade the SoftCon to SupCon in (2), the performance drops as expected, while still better than contrastive loss alone. These results verify the effectiveness of using existing free annotations to boost EO pretraining. Additionally, we ablate the weighting parameter λ in (4) and suggest a best value of 0.1. Note all results in this ablation are without continual pretraining, which we will ablate next.

2) Continual Pretraining: We ablate the continual pretraining strategy in Table VIII, which verifies the benefits of loading vision foundation models instead of pretraining from scratch. Also, we compare the performance of different vision models from ImageNet supervised weights to modern self-supervised weights. Interestingly, the stronger the vision model, the continual pretraining performance is also better. We report ImageNet top-1 linear scores of the corresponding models as a reference, which are well aligned with the cross-domain continual pretraining results.

In addition, we empirically find parameter-efficient fine-tuning (PEFT) techniques, such as BitFit [81], prompt tuning [82], and low-rank adapter (LoRA) [83], can be used for parameter-efficient continual pretraining, while not yet reaching a close performance as continually training all parameters. We report such preliminary results in the Appendix.

3) Siamese Masking: Finally, we ablate the Siamese masking strategy introduced mainly to boost training efficiency. As can be seen from Table IX, masking 50% of the patches can save almost half of the GPU memory without much performance drop. Interestingly, masking 20% of the patches can lead to even better results than no masking. This suggests that MAE-like random masking can be seen as an effective data augmentation strategy in contrastive learning. From

TABLE VII
ABLATION STUDY ON THE SOFTCON LOSS AND THE WEIGHTING PARAMETER

	BE-1%	BE-10%
Contrast only	78.9	82.1
SoftCon only	76.0	80.4
Contrast + SupCon	79.3	83.0
Contrast + SoftCon	**79.8**	**83.6**

	BE-1%	BE-10%
$\lambda = 0.01$	79.3	82.9
$\lambda = 0.1$	**79.8**	**83.6**
$\lambda = 0.5$	79.6	83.4
$\lambda = 1.0$	79.5	83.3

TABLE VIII
ABLATION STUDY ON THE SOURCE MODELS OF CONTINUAL PRETRAINING. *cont.* INDICATES CONTINUAL PRETRAINING

	Backbone	BE-1%	BE-10%	ImageNet
w/o cont.	RN50	79.8	83.6	-
cont. (ImageNet)	RN50	80.2	83.9	-
cont. (MoCov3 [56])	RN50	80.9	84.2	74.6
cont. (DINO [60])	RN50	**81.4**	**84.8**	75.3
cont. (DINOv2 [16])	ViT-S	**82.6**	**85.0**	81.1

TABLE IX
ABLATION STUDY ON THE SIAMESE MASKING RATIO

	GPU memory	BE-1%	BE-10%
w/o masking	43G	82.3	84.5
masking (20%)	36G	**82.6**	**85.0**
masking (50%)	25G	81.7	84.4

TABLE X
PERFORMANCE GAINS OF DIFFERENT COMPONENTS UNDER LINEAR PROBING ON BIGEARTHNET-10%. THE EXPLICITLY INTRODUCED COMPONENTS ARE HIGHLIGHTED IN **BOLD**

Method	Backbone	Dataset	Performance gain
Contrast	RN50	SSL4EO-S12	-
(+ SupCon)	RN50	**+ multi-label**	+0.9
+ SoftCon	RN50	-	+0.6
+ cont.	RN50	(+ ImageNet)	+1.2
+ mask	ViT-S	-	+0.5

TABLE XI
BENCHMARK RESULTS OF SSL4EO-S12-ML DATASET. WE USE 10% OF TRAINING DATA

	mAP (micro)	mAP (macro)
rand. init.	93.7	84.4
supervised	98.2	94.7
SSL4EO (linear) [10]	95.8	88.7
SoftCon (linear)	96.5	90.2
SoftCon (fine-tune)	98.8	96.1

another perspective, this design implicitly introduces the idea of masked image modeling that the model needs to know what information the masked patches have according to the unmasked branch, after which it knows the encoded features are similar. This could potentially also be one reason why the ideal masking ratio is much smaller compared to MAE, as the implicit masked image modeling plus the contrastive learning is a more challenging optimization task.

In summary, Table X provides an overview of the performance gains of each of our proposed components. It is worth noting that when introducing the multilabel annotation, the SupCon method is implicitly introduced, which can be improved by our proposed SoftCon as shown by the ablation in Table VII. Furthermore, when using the continual pretraining method, the vision dataset (e.g., ImageNet) is implicitly used for the vision model weights.

VII. CONCLUSION

This work revisits two important free resources beyond pure satellite imagery for efficient EO pretraining: open land-cover products and open vision foundation models. We build a large-scale multilabel dataset by matching an unsupervised pretraining dataset SSL4EO-S12 with Dynamic World land-cover maps. To effectively utilize the multilabel annotations, we propose a novel SoftCon method that allows soft matching between images with different label distributions. Meanwhile, we introduce strong vision models such as DINO and DINOv2 to a simple but flexible continual pretraining framework with Siamese masking. We efficiently train multispectral and SAR foundation models of both CNN and Transformer backbones that achieve SOTA performances in 10 out of 11 downstream tasks.

There are two main limitations of this work. First, our models target mainly medium-resolution data, while high-resolution images with more fine-grained semantic information remain to be explored. Second, retraining vision foundation models without any constraints inevitably makes the model forget knowledge from the source vision domain. Future work will explore more effective and flexible multimodal continual pretraining methods to build strong cross-domain cross-modal foundation models.

APPENDIX A
SSL4EO-S12-ML DATASET

A. General Information

SSL4EO-S12-ML dataset is a large-scale multilabel land-cover-land-use classification dataset. It consists of 780 371 multispectral Sentinel-2 images with size 264 × 264, divided into 247 377 non-overlapping scenes each with 1–4 multiseasonal patches. Each image has a multilabel annotation from one or more categories in nine land-cover-land-use classes.

1) Data Source: The Sentinel-2 images are from SSL4EO-S12 [10], a multimodal multitemporal dataset specifically designed for large-scale self-supervised learning. The dataset consists of four-seasonal Sentinel-1/2 SAR-optical images from 251 079 non-overlapping locations in the world, covering a wide range of geographical and temporal diversities. The multilabel annotations are derived from Dynamic World [6], a near-real-time dataset that provides continuous pixel-level land-cover monitoring in nine semantic classes. The Dynamic

TABLE XII

LINEAR PROBING MAP SCORES ON BIGEARTHNET-10% WITH DIFFERENT PRETRAINING STRATEGIES. WE REPORT PRETRAINING WITH FULL SSL4EO-S12 DATA AND 10% DATA. PEFT INDICATES DIRECTLY TRANSFERRING THE DINOV2 MODEL ON BIGEARTHNET WITH PEFT TECHNIQUES. *: SOFTCON REPRESENTS OUR SOFTCON-STYLE FULLY CONTINUAL PRETRAINING USED IN THE MAIN PAPER

	pretrain module	# pretrain params.	pretrain 100%	pretrain 10%
rand. init.	-	-	64.4	-
supervised	-	-	74.7	-
PEFT (bias)	-	-	69.2	-
PEFT (lora)	-	-	69.2	-
pretrain from scratch	all	305M	79.2	72.8
full CP (SoftCon*)	all	305M	**81.0**	**78.1**
PECP (base)	patch embed	2.6M	70.6	69.1
PECP (bias)	patch embed + bias	2.9M	72.7	70.2
PECP (prompt)	patch embed + prompt	3.0M	74.0	72.1
PECP (lora)	patch embed + lora	5.8M	74.2	73.9

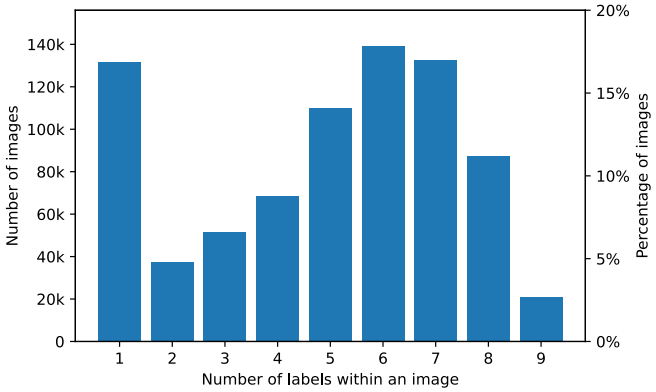


Fig. 7. Distribution of label numbers within each image.

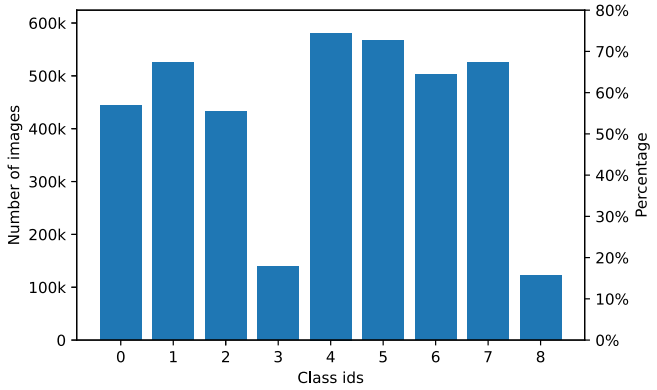


Fig. 8. Distribution of image numbers for different classes.

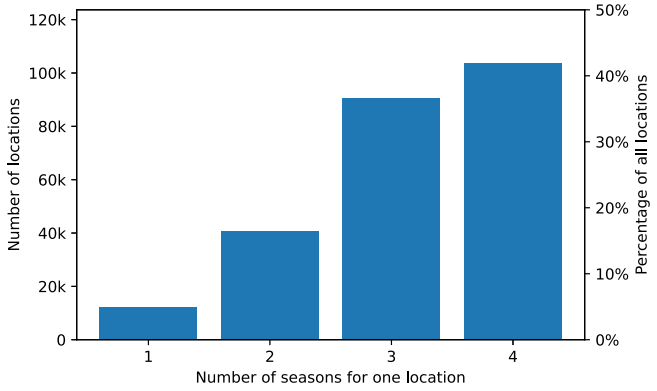


Fig. 9. Distribution of season numbers for each location.

World segmentation maps are automatically generated by algorithms developed on high-quality training data. We integrate the noisy maps into rather accurate scene-level classification labels.

2) Dataset Curation: Based on the geospatial coordinates and acquisition time, we match each SSL4EO-S12 L1C multispectral image with its corresponding Dynamic World land-cover map in Google Earth Engine. Due to the effect of clouds, a few images do not have a corresponding segmentation map, which are then dropped in our final dataset. In total, there are 247 377 scenes with a successful match for at least one season. Then, we gather the pixel labels into scene-level multilabels for each image, resulting in 780 371 labeled individual images. The workflow has been shown in Fig. 2.

B. Dataset Statistics

1) Label Number Distribution: Fig. 7 shows the distribution of the number of labels within each image. About 17% images contain a single label, while 70% images contain four or more labels.

2) Class Distribution: Fig. 8 shows the distribution of the number of images for each class. The number of images is well balanced in seven common classes, while flooded vegetation and snow/ice are less represented.

3) Season Distribution: Fig. 9 shows the distribution of the number of seasonal patches for each location. More than 40% locations have all four seasons, and more than 95% locations have at least two seasons.

C. Benchmark

We provide a preliminary benchmark on SSL4EO-S12-ML as a supervised multilabel classification task in Table XI. We split the dataset into 80% training data and 20% testing data according to the non-overlapping locations, and report micro and macro mAP on the test split. We use 10% of the training data and the full testing data for the benchmark. The table shows that this multilabel classification task is rather easy to solve, with a supervised micro mAP reaching 98.2%.

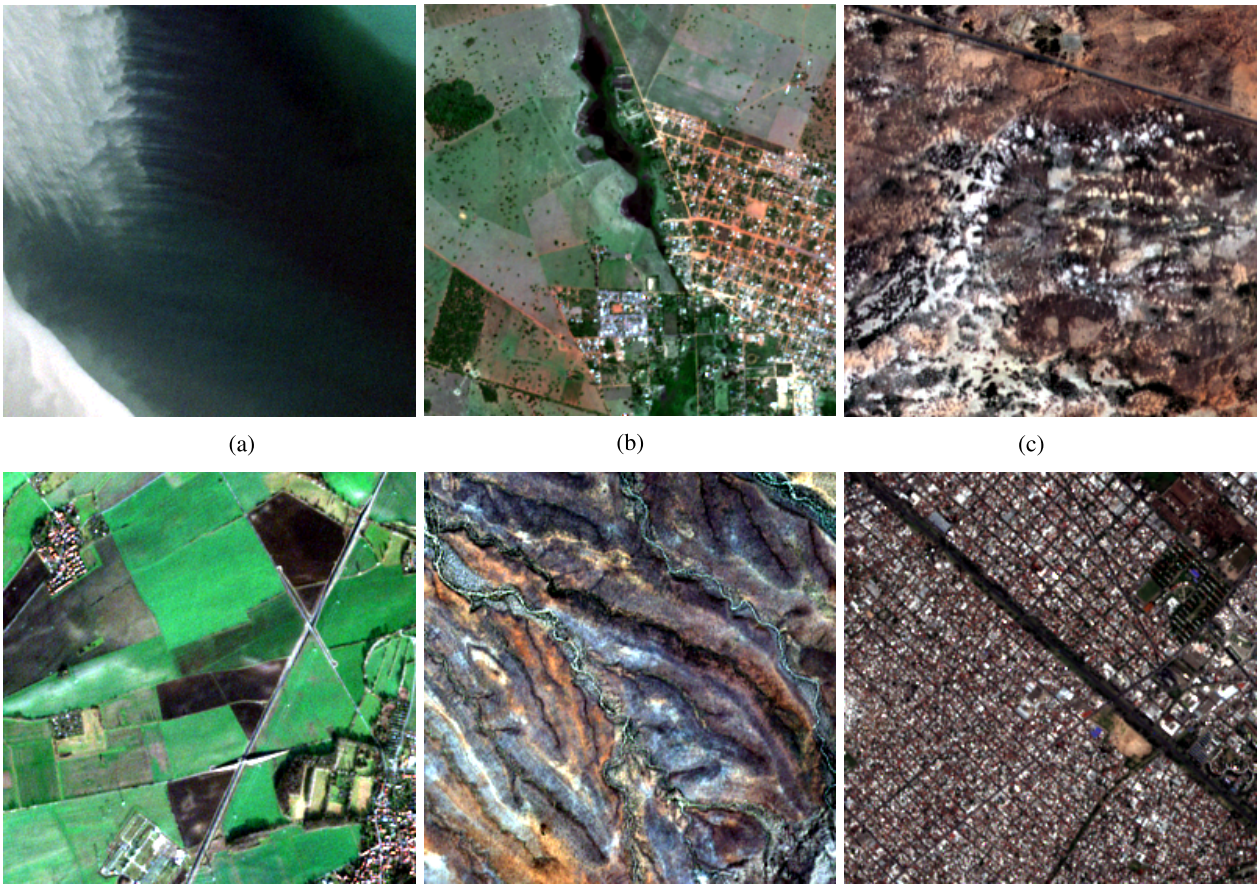


Fig. 10. SSL4EO-S12-ML example images with multilabel annotations. (a) Water. (b) Water, trees, grass, flooded vegetation, crops, shrub and scrub, built. (c) Crops, shrub and scrub, bare. (d) Water, trees, grass, crops, shrub and scrub, built, bare. (e) Grass, crops, shrub and scrub, bare. (f) Trees, grass, shrub and scrub, built, bare.

TABLE XIII
COMPUTE AND PRETRAINING TIME

Modality	Backbone	GPUs	Training time
MS	RN50	4xA100	21h
	ViT-S/14	4xA100	25h
	ViT-B/14	8xA100	15h
SAR	RN50	4xA100	7h
	ViT-S/14	4xA100	8h
	ViT-B/14	8xA100	7h

This is similar to the training metric, indicating the balanced dataset split and rather good label quality. In addition, we also test the dataset as a downstream task to evaluate pretrained models. As is shown in the table, consistent performances as other popular datasets in the main paper are observed: 1) pretrained models improve upon random initialization and 2) SoftCon outperforms existing models such as SSL4EO. Therefore, SSL4EO-S12-ML can be considered as a global multilabel classification benchmark dataset that complements existing datasets such as BigEarthNet [53], which covers only Europe.

D. Data Examples

Fig. 10 provides some example images and corresponding multilabels in the SSL4EO-S12-ML dataset covering different landscapes.

TABLE XIV
DFC2020 FINE-TUNING HYPERPARAMETERS

	DFC2020 (ResNet)	DFC2020 (ViT)
Backbone	ResNet50	ViT-B/14, ViT-S/14
Input size	224x224	224x224
Augmentation	ResizedCrop (0.5,2.0), HorizontalFlip, VerticalFlip	ResizedCrop (0.5,2.0), HorizontalFlip, VerticalFlip
Batch size	16	16
Learning rate	1e-3	5e-3
LR schedule	poly	poly
Optimizer	SGD	AdamW
Weight decay	5.00E-04	5.00E-02
Warm-up	0	1000
Iter	20K	40K
Head	DeepLabv3+	UperNet

APPENDIX B
PARAMETER EFFICIENT CONTINUAL PRETRAINING

Besides SoftCon weight initialization, we also explored other continual pretraining strategies, such as the adaptation of PEFT techniques. PEFT is usually used in fine-tuning foundation models to specific downstream tasks with low cost. We studied the feasibility of adapting it to continual pretraining, termed parameter efficient continual pretraining (PECP), to build foundation models for a target domain such as EO.

TABLE XV
LINEAR PROBING HYPERPARAMETERS ON BIGEARTHNET, EUROSAT, AND FMOW-S2

	BigEarthNet	EuroSAT	fMoW-S2
Backbone	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14
Input size	224x224	224x224	224x224
Augmentation	ResizedCrop (0.8,1.0), HorizontalFlip	ResizedCrop (0.2,1.0), HorizontalFlip	ResizedCrop (0.2,1.0), HorizontalFlip
Batch size	256	256	1024
Learning rate	0.1	1e-3	4e-4
LR schedule	cos	step	cos
Optimizer	SGD	SGD	AdamW
Weight decay	0	0	0.01
Warm-up	0	0	0
Epoch	100	100	100
Feature block	4	1	4

TABLE XVI
FINE-TUNING HYPERPARAMETERS ON BIGEARTHNET, EUROSAT, AND FMOW-S2

	BigEarthNet	EuroSAT	fMoW-S2
Backbone	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14
Input size	224x224	224x224	224x224
Augmentation	ResizedCrop (0.8,1.0), HorizontalFlip, VerticalFlip	ResizedCrop (0.2,1.0), HorizontalFlip	ResizedCrop (0.2,1.0), HorizontalFlip, mixup&cutmix
Batch size	256	256	1024
Learning rate	1e-4	1e-3	4e-4
LR schedule	cos	step	cos
Optimizer	AdamW	SGD	AdamW
Weight decay	0.01	0	0.05
Warm-up	0	0	0
Epoch	50	100	50

TABLE XVII
RESNET LINEAR PROBING (LEFT) AND FINE-TUNING (RIGHT) HYPERPARAMETERS ON BIGEARTHNET AND EUROSAT

	BigEarthNet	EuroSAT
Backbone	RN50	RN50
Input size	224x224	224x224
Augmentation	ResizedCrop (0.8,1.0), HorizontalFlip	ResizedCrop (0.2,1.0), HorizontalFlip
Batch size	256	256
Learning rate	8	8
LR schedule	step	step
Optimizer	SGD	SGD
Weight decay	0	0
Warm-up	0	0
Epoch	100	100

	BigEarthNet	EuroSAT
Backbone	RN50	RN50
Input size	224x224	224x224
Augmentation	ResizedCrop (0.8,1.0), HorizontalFlip, VerticalFlip	ResizedCrop (0.2,1.0), HorizontalFlip
Batch size	256	256
Learning rate	1e-3	0.1
LR schedule	cos	step
Optimizer	AdamW	SGD
Weight decay	0.01	0
Warm-up	0	0
Epoch	50	100

We study PECP with three representative PEFT techniques: bias-tuning like BiTFiT [81], visual prompt tuning [82], and LoRA [83]. To fully evaluate the capacity of large foundation models, we transfer DINOv2 ViT-Large with about 300M parameters which is on par with commonly studied models in PEFT. To save computing costs and to rule out other factors, we use a simple MAE [15] for pretraining.

Table XII presents linear probing results on BigEarthNet with different pretraining strategies. First, as the table shows, straightforward PEFT from DINOv2 does provide benefits compared to random frozen encoder. However, such benefits are rather limited compared to in-domain pretraining. Second, compared to pretraining from scratch, SoftCon-style continual pretraining offers significant improvement, especially when the in-domain data is limited in size: continual pretraining on 10% data is comparable to pretraining full data from scratch. Third, all PECP strategies provide reasonable benefits with only a small fraction of parameters trainable, outperforming direct PEFT. Among them, LoRA performs slighter better than others. Similar to full continual pretraining, PECP also remains the performance when the size of pretraining data is restricted. Lastly, all the simple PECP strategies are not close to full continual pretraining, restricting the practical usage. This preliminary study motivates us to use full continual pretraining for SoftCon in the main papaer. However, with more research toward advanced designs, we believe PECP bears potential for future work, as the flexible adapters can pave the way toward unified cross-domain foundation models.

APPENDIX C
IMPLEMENTATION DETAILS

A. Pretraining

We pretrain SoftCon with ResNet [63] and ViT [64] backbones on the proposed multilabel dataset SSL4EO-S12-ML, which consists of 247 377 scenes with 1–4 seasons. We normalize the 16-bit images to 8-bit with the mean and standard

TABLE XVIII
TRANSFER LEARNING HYPERPARAMETERS ON THE FOUR GEO-BENCH DATASETS

	m-so2sat	m-brick-kiln	m-cashew-plantation	m-SA-crop-type
Backbone	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14	ViT-B/14, ViT-S/14
Input size	224x224	224x224	224x224	224x224
Augmentation	ResizedCrop (0.8,1.0), HorizontalFlip	ResizedCrop (0.8,1.0), HorizontalFlip	ResizedCrop (0.6,1.0), HorizontalFlip, VerticalFlip, Rotate	ResizedCrop (0.6,1.0), HorizontalFlip, VerticalFlip, Rotate
Batch size	256	256	64	64
Learning rate	1.0	10	0.01	0.01
LR schedule	cos	cos	cos	cos
Optimizer	LARS	LARS	AdamW	AdamW
Weight decay	0	0	0.01	0.01
Warm-up	0	0	3	3
Epoch	50	50	50	50

deviation provided in [10]. If there are multiple seasons for one scene, we randomly choose two for the base encoder and the momentum encoder, respectively. Data augmentations follow [10], including random crop (to the size 224 × 224), color jitter, grayscale, Gaussian blur, and random flip.

We adapt MoCo-v2 [55] for ResNet50 and MoCo-v3 [56] for ViT-S/14 and ViT-B/14, with two separate projectors to get embeddings for the Contrast and SoftCon loss, respectively. Each projector consists of two linear layers and one ReLU activation function in between. The weighting parameter λ trading off the two losses is 0.1. We set a queue size of 16 384 with a batch size of 256 for MoCo-v2, and a batch size of 1024 for MoCo-v3. For MoCo-v2, the learning rate starts from 0.03, followed by cosine decay to 0; for MoCo-v3, the learning rate is warmed up to $1.5e^{-4}$ for 10 epochs, followed by cosine decay to 0. The optimizer is SGD for MoCo-v2 and AdamW for MoCo-v3. The softmax temperature is 0.2 for both MoCo-v2 and MoCo-v3.

We load ResNet50 weights from DINO [60] and ViT-S/14 and ViT-B/14 weights (without register) from DINOv2 [16] and conduct continual pretraining for 100 epochs. For ViTs, we randomly mask out 20% patches and send the remaining patches to the trainable encoder. All patches without masking are sent to the momentum encoder. Training is distributed in two nodes each with four NVIDIA A100 GPUs. Detailed compute and training time for different backbones are shown in Table XIII.

B. Downstream Tasks

1) Downstream Tasks: We evaluate the pretrained backbones by linear probing and fine-tuning in eight downstream tasks, including four land-cover-land-use classification/segmentation datasets: BigEarthNet [53], EuroSAT [66], fMoW-sentinel [24] and DFC2020 [67], and four multispectral datasets covering different applications from GEO-Bench [69]: m-so2sat, m-brick-kiln, m-cashew-plantation, and m-SA-crop-type.

For classification datasets, we conduct linear probing and fine-tuning; for DFC2020, we fine-tune DeepLabv3+ [77] for ResNet backbone and UperNet [78] for ViT backbones; for GEO-Bench segmentation datasets, we freeze the encoder and train a UperNet decoder. For linear probing with ViT backbones, we pick either the output features from the last block, or the concatenation of features from the last four blocks following DINOv2 [16]. We do a simple grid search for the learning rate for each dataset. Specific hyperparameters for each dataset are summarized in Tables XIV–XVIII.

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